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LAB REPORT
on

Artificial Intelligence (23CS5PCAIN)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the Lab work entitled "Artificial Intelligence (23CS5PCAIN)" carried out by **ANOOP Y PATIL(1BM23CS042),** who is a bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Artificial Intelligence (23CS5PCAIN) work prescribed for the said degree.

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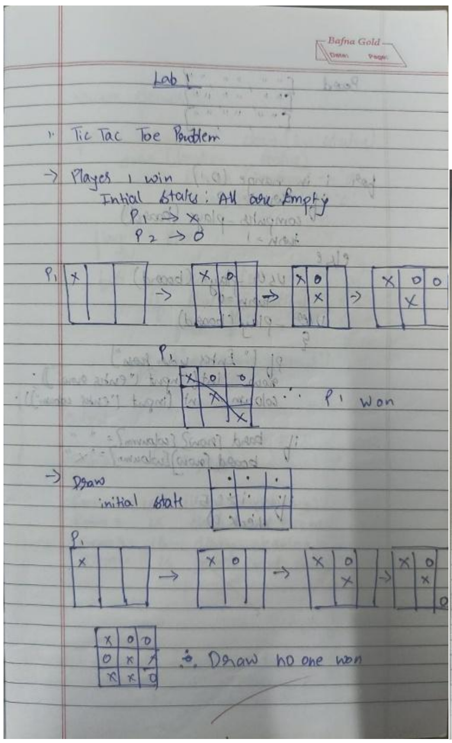
Github Link:
https://github.com/anoop042/AI-LAB

PROGRAM 1:

Implement Tic –Tac –Toe Game
Implement vacuum cleaner agent

ALGORITHM:



Lab 1

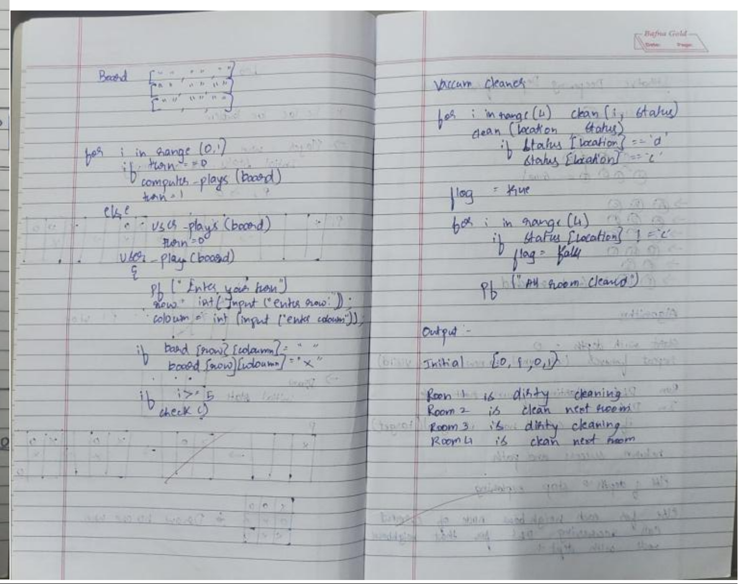
1. Tic Tac Toe Problem

→ Player 1 win
Initial State: All are Empty
$P_1 \to x$
$P_2 \to 0$

∴ P1 won

→ Draw
initial state

∴ Draw no one won



Board

for i in range (0,?)
if turn == 0
computer_plays (board)
turn = 1
else
user_plays (board)
turn = 0
user_play (board)
{
pf ("Enter your turn")
row = int(input ("enter row"))
column = int (input ("enter column"))
if board [row][column] == " "
board [row][column] = "x"
if i >= 5
check ()

Vaccum cleaner
for i in range (4) clean (i, status)
clean (location, status)
if status [location] == 'd'
status [location] = 'c'
flag = true
for i in range (4)
if status [location] != 'c'
flag = false
pf ("All room cleaned")

Output :-
Initial [0, 1, 0, 1]
Room 1 is dirty cleaning
Room 2 is clean next room
Room 3 is dirty cleaning
Room 4 is clean next room

CODE:

```
#TicTacToe
import math
def check_win(boar
wins = [(0,1,2),(3,4,5),(6,7,8),


        (0,3,6),(1,4,7),(2,5,8),
        (0,4,8),(2,4,6)]
    for a,b,c in wins:
        if board[a] == board[b] == board[c] != 0:
            return board[a]
    return 0 if 0 not in board else None

def selection(board, player):
    result = check_win(board)
    if result is not None: return result
    scores = []
    for i in range(9): if
        board[i] == 0:
            board[i] = player
            scores.append(selection(board, -player))
            board[i] = 0
    return max(scores) if player == 1 else min(scores)

def best_move(board):
    return max((selection(board[:i]+[1]+board[i+1:], -1), i)
               for i in range(9) if board[i]==0)[1]

def print_board(board):
    s = {1:'X', -1:'O', 0:' '}
    for i in range(0,9,3):
        print(f"{s[board[i]]}|{s[board[i+1]]}|{s[board[i+2]]}"
        ) if i<6: print(" ")

board=[0]*9; player=1
while True:
    print_board(board)
    res = check_win(board)
    if res is not None:
        print("X wins!" if res==1 else "O wins!" if res==-1 else "Tie!")
        break
    if player==1:
        print("AI thinking. ")
        board[best_move(board)] = 1
    else:
        move=int(input("Move (0-8): "))
        if 0<=move<=8 and board[move]==0: board[move]=-1
        else: print("Invalid"); continue
    player*=-1
```

```
#Vaccum Cleaner
 def isclean(list,place):
        if(list[place]==1):
          list[place] = 0
        return False
        else:
          return True

      list = [1,1]
      place = 0
      count = 2

      while(count!=0):
        if(place==0):
        if(isclean(list,place)):
            print("Location A is already cleaned")
          else:
            print("Location A is not clean")
            print("cleaning Location A")
            print("Locatin A has been cleaned")

          count = count-1
          place = 1

        if(isclean(list,place)):
          print("Location B is already cleaned")
        else:
      print("Location B is not clean") print("cleaning Location B") print("Locatin B has been cleaned") count
      = count-1
```

OUTPUT:

```
 0 | 1 | 2
---+---+---
 3 | 4 | 5
---+---+---
 6 | 7 | 8
AI thinking...
 0 | 1 | 2
---+---+---
 3 | 4 | 5
---+---+---
 6 | 7 | X
Your move (0-8): 2
 0 | 1 | O
---+---+---
 3 | 4 | 5
---+---+---
 6 | 7 | X
AI thinking...
 0 | 1 | O
---+---+---
 3 | 4 | 5
---+---+---
 6 | X | X
Your move (0-8): 6
 0 | 1 | O
---+---+---
 3 | 4 | 5
---+---+---
 O | X | X
AI thinking...
 0 | 1 | O
---+---+---
 3 | X | 5
---+---+---
 O | X | X
Your move (0-8): 0
 O | 1 | O
---+---+---
 3 | X | 5
```

```
Your move (0-8): 0
 O | 1 | O
---+---+---
 3 | X | 5
---+---+---
 O | X | X
AI thinking...
 O | X | O
---+---+---
 3 | X | 5
---+---+---
 O | X | X
X wins!
```

```
Location A is not clean
cleaning Location A
Locatin A has been cleaned
Location B is not clean
cleaning Location B
Locatin B has been cleaned
```

PROGRAM 2:

Implement 8 puzzle problems using Depth First Search (DFS)
Implement Iterative deepening search algorithm

ALGORITHM:

8-puzzel problem
using iterative

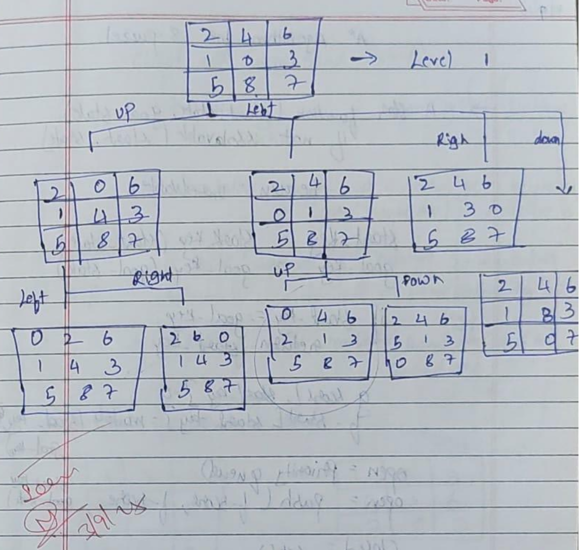
Algorithm

- Taking the input of the 2D array
- Initialise the depth as 0
- Check the row no. of 0
- Define a goal state [1,2,3,4,5,6,7,8,0]

→ Start with intial puzzel state
→ Set depth limit as 0
→ Repeat the following until you get the goal state.
- Peform DLS from initial state & with current depth limit
- In DLS current state equal to goal state stop & return the solution path.
- If current depth is 0 return failure.

→ Else for each possible value from current state
- recursively apply DLS to new state by decreased depth by 1
- If it return a solution return the path with solution.

* If no solution is found at current depth limit, increase by 1 & repeat. & when goal is found output the sequence of moves from start to goal.



CODE:

```
goal = [[1,2,3],
        [8,0,4],
        [7,6,5]]

moves = [(-1,0),(1,0),(0,-1),(0,1)]

def find_zero(s):
    for i in range(3):
        for j in range(3):
            if s[i][j] == 0:
                return i, j


def get_neighbors(s):
    x, y = find_zero(s)
    out = []
    for dx, dy in moves:
        nx, ny = x+dx, y+dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            t = [r[:] for r in s]
            t[x][y], t[nx][ny] = t[nx][ny], t[x][y]
            out.append(t)
    return out


def dfs(s, visited):
    if s == goal:
        return [s]
    visited.add(str(s))
    for nxt in get_neighbors(s):
```

```
        if str(nxt) not in visited:
            p = dfs(nxt, visited)
            if p:
                return [s] + p
    return None


def dls(s, goal, depth, path, visited):
    if s == goal:
        return path
    if depth == 0:
        return None
    visited.add(str(s))
    for nxt in get_neighbors(s):
        if str(nxt) not in visited:
            res = dls(nxt, goal, depth-1, path+[nxt], visited)
            if res:
                return res
    return None


def ids(start, goal, limit=20):
    for d in range(limit+1):
        visited = set()
        r = dls(start, goal, d, [start], visited)
        if r:
            return r
    return None



if name == "main":

start = [[1,2,3], [8,6,4], [0,7,5]]



print("DFS:")

    sol1 = dfs(start, set())

    if sol1:

        print("Moves:", len(sol1)-1)

        for st in sol1:

            for r in st:

                print(r)

            print("-"*40)

    else:
```

```
        print("No solution.")


print("\nIDS:")

    sol2 = ids(start, goal, 20)
    if sol2:

        print("Moves:", len(sol2)-1)

        for st in sol2:

            for r in st:

                print(r)

            print("-"*40)

    else:

        print("No solution.")
```

Output:

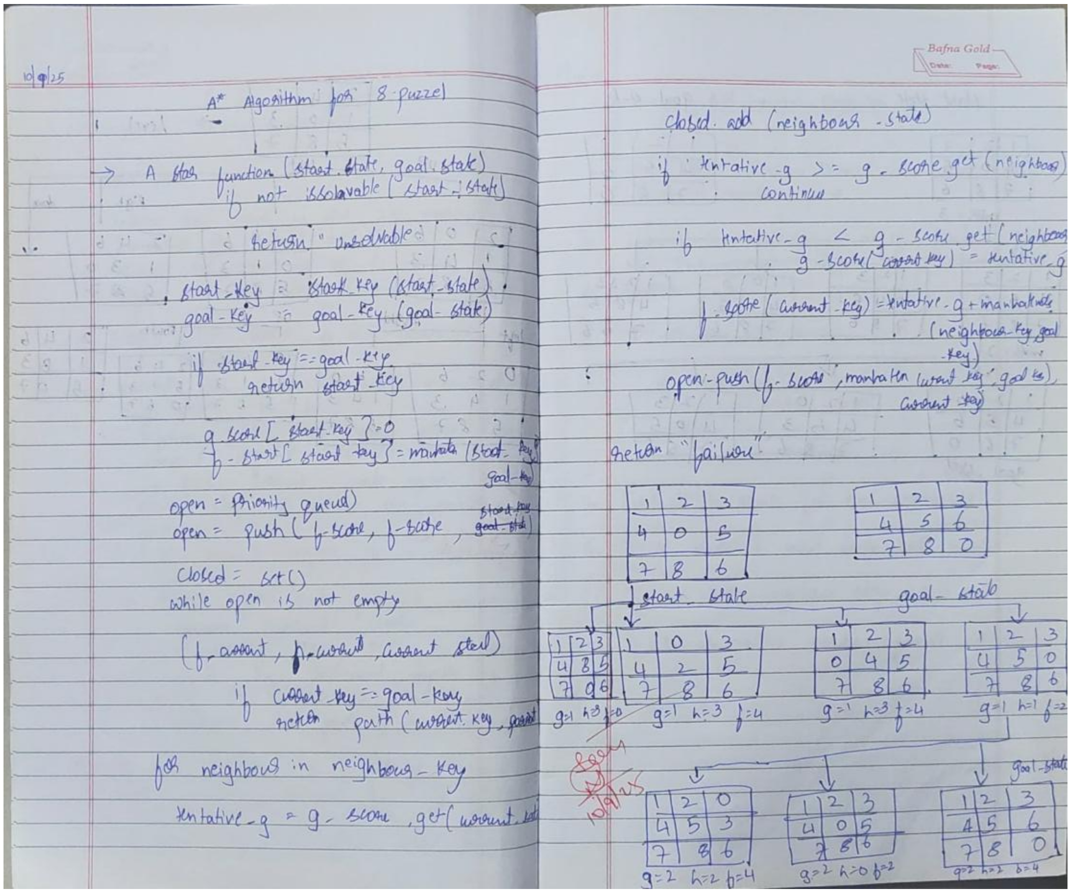
```
DFS:
Moves: 20
[1, 2, 3]
[8, 6, 4]
[0, 7, 5]
----------------------------------------
[1, 2, 3]
[0, 6, 4]
[8, 7, 5]
----------------------------------------
[0, 2, 3]
[1, 6, 4]
[8, 7, 5]
----------------------------------------
[2, 0, 3]
[1, 6, 4]
[8, 7, 5]
----------------------------------------
[2, 6, 3]
[1, 0, 4]
[8, 7, 5]
----------------------------------------
[2, 6, 3]
[1, 7, 4]
[8, 0, 5]
----------------------------------------
[2, 6, 3]
[1, 7, 4]
[0, 8, 5]
```

```
----------------------------------------
[2, 6, 3]
[0, 7, 4]
[1, 8, 5]
----------------------------------------
[0, 6, 3]
[2, 7, 4]
[1, 8, 5]
----------------------------------------
[6, 0, 3]
[2, 7, 4]
[1, 8, 5]
----------------------------------------
[6, 7, 3]
[2, 0, 4]
[1, 8, 5]
----------------------------------------
[6, 7, 3]
[2, 8, 4]
[1, 0, 5]
----------------------------------------
[6, 7, 3]
[2, 8, 4]
[0, 1, 5]
----------------------------------------
[6, 7, 3]
[0, 8, 4]
[2, 1, 5]
----------------------------------------
[0, 7, 3]
[6, 8, 4]
[2, 1, 5]
```

PROGRAM 3:

Implement A* search algorithm

ALGORITHM:



10/9/25

A* Algorithm for 8-puzzel

→ A star function (start.state, goal.state)
    if not issolavable (start_state)
        return "unsolvable"
    start_key = start_key (start_state)
    goal_key = goal_key (goal_state)
    if start_key == goal_key
        return start_key
    g_score[start_key] = 0
    f_start[start_key] = manhatan (start_key, goal_key)
open = priority queue()
open = push (f_score, f_score, start_key)
closed = set()
while open is not empty
    (f_current, f_current, current state)
        if current_key == goal_key
        return path (current.key, parent
for neighbour in neighbour_key
    tentative_g = g_score.get(current_state
    closed.add (neighbour_state)
    if tentative_g >= g_score.get (neighbour)
        continue
    if tentative_g < g_score.get (neighbour
        g_score(current_key) = tentative_g
        f_score (current_key) = tentative_g + manhattan (neighbour_key, goal_key)
    open.push (f_score, manhattan (current_key, goal_key), current_key)
return "failure"

start - state

1	2	3
4	0	5
7	8	6

goal - state

1	2	3
4	5	6
7	8	0

miss = 3

1	2	3
4	5	0
7	8	6

1	2	3
0	4	5
7	8	6

1	0	3
4	2	5
7	8	6

1	2	3
4	8	5
7	0	6

miss = 2

1	2	3
4	5	6
7	8	0

goal - state

1	2	0
4	5	3
7	8	6

1	2	3
4	0	5
7	8	6

CODE:

```
import heapq
class PuzzleState:
    def init (self, board, parent=None, move="", depth=0, cost=0):
        self.board = board
        self.parent = parent
        self.move = move
        self.depth = depth
        self.cost = cost

    def lt (self, other):
        return self.cost < other.cost

    def blank_pos(self):
        return self.board.index(0)

    def expand(self):
        b = self.blank_pos()
        row, col = divmod(b, 3)
        dirs = {
            "Up": (row - 1, col),
            "Down": (row + 1, col),
            "Left": (row, col - 1),
            "Right": (row, col + 1)
        }
        nxt = []
        for mv, (r, c) in dirs.items():
            if 0 <= r < 3 and 0 <= c < 3:
                idx = r * 3 + c

nb = self.board[:]
                nb[b], nb[idx] = nb[idx], nb[b]
                nxt.append(PuzzleState(nb, self, mv, self.depth + 1))
        return nxt

    def build_path(self):
        p, node = [], self
        while node:
            p.append((node.move, node.board, node.depth))
            node = node.parent
        return list(reversed(p))


def misplaced_tiles(state, goal):
    return sum(1 for i in range(9) if state.board[i] not in (0, goal[i]))
def manhattan_distance(state, goal):
    d = 0
    for i, v in enumerate(state.board):
        if v != 0:
            r1, c1 = divmod(i, 3)
```

```
        r2, c2 = divmod(goal.index(v), 3)
        d += abs(r1 - r2) + abs(c1 - c2)
    return d

def a_star(start, goal, h):
    opened = []
    closed = set()
    nodes = 0
    s = PuzzleState(start)
    s.cost = h(s, goal)
    heapq.heappush(opened, s)

    while opened:
        cur = heapq.heappop(opened)
        nodes += 1

        if cur.board == goal:
            return cur.build_path(), nodes

        closed.add(tuple(cur.board))

        for nxt in cur.expand():
            if tuple(nxt.board) in closed:
                continue
            nxt.cost = nxt.depth + h(nxt, goal)
            heapq.heappush(opened, nxt)

    return None, nodes


def print_solution(path, total_nodes):
    print("Steps:\n")
    for mv, st, d in path:
```

```
        label = "Start" if mv == "" else f"Move {mv}"
        print(f"{label} | Depth {d}")
        for i in range(0, 9, 3):
            print(" ".join(str(x) if x != 0 else " " for x in st[i:i+3]))
        print()
    print(f"Total Moves: {len(path)-1}")
    print(f"Nodes Expanded: {total_nodes}")


if __name__ == "__main__":
    start = [1, 2, 3,
             4, 0, 6,
             7, 5, 8]

    goal = [1, 2, 3,
            4, 5, 6,
            7, 8, 0]

    print("A* (Misplaced Tiles)\n")
    sol1, n1 = a_star(start, goal, misplaced_tiles)
    if sol1:
        print_solution(sol1, n1)
    else:
        print("No solution.")

    print("\nA* (Manhattan Distance)\n")
    sol2, n2 = a_star(start, goal, manhattan_distance)
    if sol2:
        print_solution(sol2, n2)
    else:
        print("No solution.")
```

OutPut:

```
A* (Manhattan Distance)

Steps:

Start | Depth 0
1 2 3
4   6
7 5 8

Move Down | Depth 1
1 2 3
4 5 6
7   8

Move Right | Depth 2
1 2 3
4 5 6
7 8

Total Moves: 2
Nodes Expanded: 3
```

```
A* (Misplaced Tiles)

Steps:

Start | Depth 0
1 2 3
4   6
7 5 8

Move Down | Depth 1
1 2 3
4 5 6
7   8

Move Right | Depth 2
1 2 3
4 5 6
7 8

Total Moves: 2
Nodes Expanded: 3
```

PROGRAM 4:

Implement Hill Climbing search algorithm to solve N-Queens problem

ALGORITHM:

8/10/25

Bafna Gold
Date: Page:

Kill climbing Algorithm

1. Start with random board one queen per coloum in random row
2. Compute heuristic (h) = no. of attacking pairs ~~i.e one queen should no attack another queen~~
3. Generating neighbours

   If board [col] != row
   new_board = board
   board [col] = row
   h = calculate_attacks
4. Evaluate h values for each neighbours
5. Choose the h value with least value.
6. If no better neighbour stop
7. If h is zero solution found

Example 4 queens

	0	1	2	3
0	Q			
1		Q		
2			Q	
3				Q

here h = 6 attacking pairs.

Now change the queen's position to find the best neighbour

	0	1	2	3
0		Q		
1	Q			
2		Q		
3				Q

h = 4

Best neighbour

	0	1	2	3
0			Q	
1	Q			
2				Q
3		Q		

h = 0

CODE:

```
import random
import time

def print_board(b):
    n = len(b)
    for i in range(n):
        row = ["Q" if b[i] == j else "." for j in range(n)]
        print(" ".join(row))
    print()

def cost(b):
    n = len(b)
    c = 0
    for i in range(n):
        for j in range(i + 1, n):
            if b[i] == b[j] or abs(b[i] - b[j]) == abs(i - j):
                c += 1
    return c

def best_neighbor(b):
    n = len(b)
    best_b = list(b)
```

```
    best_c = cost(b)
    for r in range(n):
        for c in range(n):
            if b[r] != c:
                temp = list(b)
                temp[r] = c
                k = cost(temp)
                if k < best_c:
                    best_c = k
                    best_b = temp
    return best_b, best_c

def hill(n):
    state = [random.randint(0, n - 1) for _ in range(n)]
    c = cost(state)

    print("Initial Board:")
    print_board(state)
    print("Initial Cost:", c, "\n")

    step = 1
    while True:
        print("Step", step)
        print("Current Board:")
        print_board(state)
        print("Current Cost:", c)
        time.sleep(0.2)

        nxt, nxt_c = best_neighbor(state)
        print("Best Neighbor Cost:", nxt_c, "\n")
        time.sleep(0.25)

        if nxt_c >= c:
            break

        state = nxt
        c = nxt_c
        step += 1

    print("Final Board:")
    print_board(state)
    print("Final Cost:", c)

    if c == 0:
        print("Solution Found")
    else:
        print("Local Minimum Reached")

hill(6)
```

OutPut:

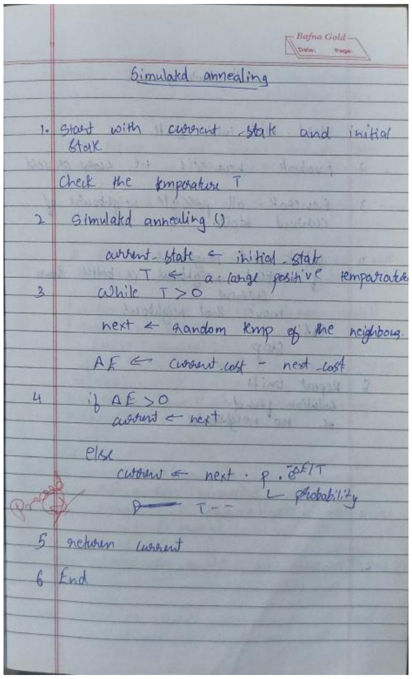
```
Initial Board:
. . . . . Q
Q . . . . .
Q . . . . .
. . . Q . .
. . . Q . .
. . Q . . .

Initial Cost: 4

Step 1
Current Board:
. . . . . Q
Q . . . . .
Q . . . . .
. . . Q . .
. . . Q . .
. . Q . . .

Current Cost: 4
Best Neighbor Cost: 2

Step 2
Current Board:
. . . . . Q
Q . . . . .
Q . . . . .
. . . Q . .
. . . . Q .
. . Q . . .

Current Cost: 2
Best Neighbor Cost: 2
```

```
Current Cost: 2
Best Neighbor Cost: 2

Final Board:
. . . . . Q
Q . . . . .
Q . . . . .
. . . Q . .
. . . . Q .
. . Q . . .

Final Cost: 2
Local Minimum Reached
```

PROGRAM 5:

Simulated Annealing to Solve 8-Queens problem

ALGORITHM:



Bafna Gold
Date: Page:

Simulated annealing

1. Start with current state and initial state

Check the temperature T

2 Simulated annealing ()

current_state ← initial_state
T ← a large positive temparature

3 While T > 0

next ← random temp of the neighbour.

ΔE ← current_cost - next_cost

4 if $\Delta E > 0$
current ← next

else
current ← next . p . $e^{-\Delta E/T}$
└ probability

T - -

5 return current

6 End

CODE:

```
import random
import math

def print_board(board):
    n = len(board)
    for i in range(n):
        row = ["Q" if board[i] == j else "." for j in range(n)]
        print(" ".join(row))
    print()

def calculate_cost(board):
    n = len(board)
    cost = 0
    for i in range(n):
        for j in range(i + 1, n):
            if board[i] == board[j] or abs(board[i] - board[j]) == abs(i - j):
                cost += 1
    return cost

def random_neighbor(board):
    n = len(board)
    neighbor = list(board)
    row = random.randint(0, n - 1)
    col = random.randint(0, n - 1)
    neighbor[row] = col
    return neighbor
```

```
def simulated_annealing(n, initial_temp=100.0, cooling_rate=0.95, stopping_temp=1.0,
max_steps=20):
    current_board = [random.randint(0, n - 1) for _ in range(n)]
    current_cost = calculate_cost(current_board)
    temperature = initial_temp
    step = 1

    print("Initial Board:")
    print_board(current_board)
    print(f"Initial Cost: {current_cost}\n")

    while temperature > stopping_temp and current_cost > 0 and step <= max_steps:
        neighbor = random_neighbor(current_board)
        neighbor_cost = calculate_cost(neighbor)
        delta = neighbor_cost - current_cost

        if delta < 0 or random.random() < math.exp(-delta / temperature):
            current_board = neighbor
            current_cost = neighbor_cost

        print(f"Step {step}: Temp={temperature:.3f}, Cost={current_cost}")
        step += 1
        temperature *= cooling_rate

    print("\nFinal Board:")
    print_board(current_board)
    print(f"Final Cost: {current_cost}")

    if current_cost == 0:
        print("Goal State Reached!")
    else:
        print("Terminated before reaching goal.")

if __name__ == "__main__":
    simulated_annealing(8, initial_temp=100.0, cooling_rate=0.95, stopping_temp=1.0,
max_steps=20)
```

OutPut:

```
Initial Board:
. . Q . . . . .
. . Q . . . . .
Q . . . . . . .
. . . Q . . . .
. Q . . . . . .
. . . . . Q . .
. . . . . . . Q
. . . . . . Q .

Initial Cost: 5

Step 1: Temp=100.000, Cost=5
Step 2: Temp=95.000, Cost=5
Step 3: Temp=90.250, Cost=5
Step 4: Temp=85.737, Cost=4
Step 5: Temp=81.451, Cost=4
Step 6: Temp=77.378, Cost=5
Step 7: Temp=73.509, Cost=5
Step 8: Temp=69.834, Cost=8
Step 9: Temp=66.342, Cost=9
Step 10: Temp=63.025, Cost=10
Step 11: Temp=59.874, Cost=10
Step 12: Temp=56.880, Cost=10
Step 13: Temp=54.036, Cost=10
Step 14: Temp=51.334, Cost=10
Step 15: Temp=48.767, Cost=10
Step 16: Temp=46.329, Cost=12
Step 17: Temp=44.013, Cost=8
Step 18: Temp=41.812, Cost=8
Step 19: Temp=39.721, Cost=9
Step 20: Temp=37.735, Cost=9
```

```
Final Board:
. . . . Q . . .
. . Q . . . . .
. . Q . . . . .
. . Q . . . . .
Q . . . . . . .
. . . . . Q . .
. Q . . . . . .
. . . . . . . Q

Final Cost: 9
Terminated before reaching goal.
```

PROGRAM 6:

Create a knowledge base using propositional logic and show that the given query entails the knowledge base or not.

ALGORITHM:

LAB -5

Q Creak a knowledge based using propositional logic and show that the given entails the knowledge based or not.

TT Entails (KB, α) return True or False
inputs: KB, the knowledge base, a sentance in propositional logic
α, the query a sentence in propositional logic.

A list of the propositional symbol in KB and α return TT_CHECK_ALL (KB,α, Symbols, {})

function TT_CHECK_ALL (KB, α, Symbols, model)
return True or False
if EMPTY ? (Symbol) :
if PL_True? (KB, model) then return PL_TRUE (α, mode)
else return True // when KB is false, always return True

Else do
P ← First (Symbols)
rest ← Rest (Symbols)
return (TT_CHECK_ALL (KB, α, rest, model ∪ { P = true })
and
(TT_CHECK_ALL (KB, α, rest, model ∪ { P = False })

Q) Q → P
P → ¬Q
Q ∨ R

i) Construct Truth Table
ii) Does KB entails R
iii) Does KB entails R→P
iv) Does KB entailes Q→P

i) Truth Table

P	Q	R	Q→P	P→¬Q	Q∨R
T	T	T	T	F	T
T	T	F	T	F	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	F

ii) Yes
Because, considering row 3 & 7 the KB Q→P, P→¬Q & Q∨R is True and R is True for both rows.

iii) No
Because Row 7 model KB is True But R→P is False

iv) Yes Because row 3 & 7 all models are True and Q→R is also True

Truth Table

A	B	C	A∨C	B∨¬C	KB
F	F	F	F	T	F
F	F	T	T	F	F
F	T	F	F	T	F
F	T	T	T	T	T
T	F	F	T	T	T
T	F	T	T	F	F
T	T	F	T	T	T
T	T	T	T	T	T

Entailment

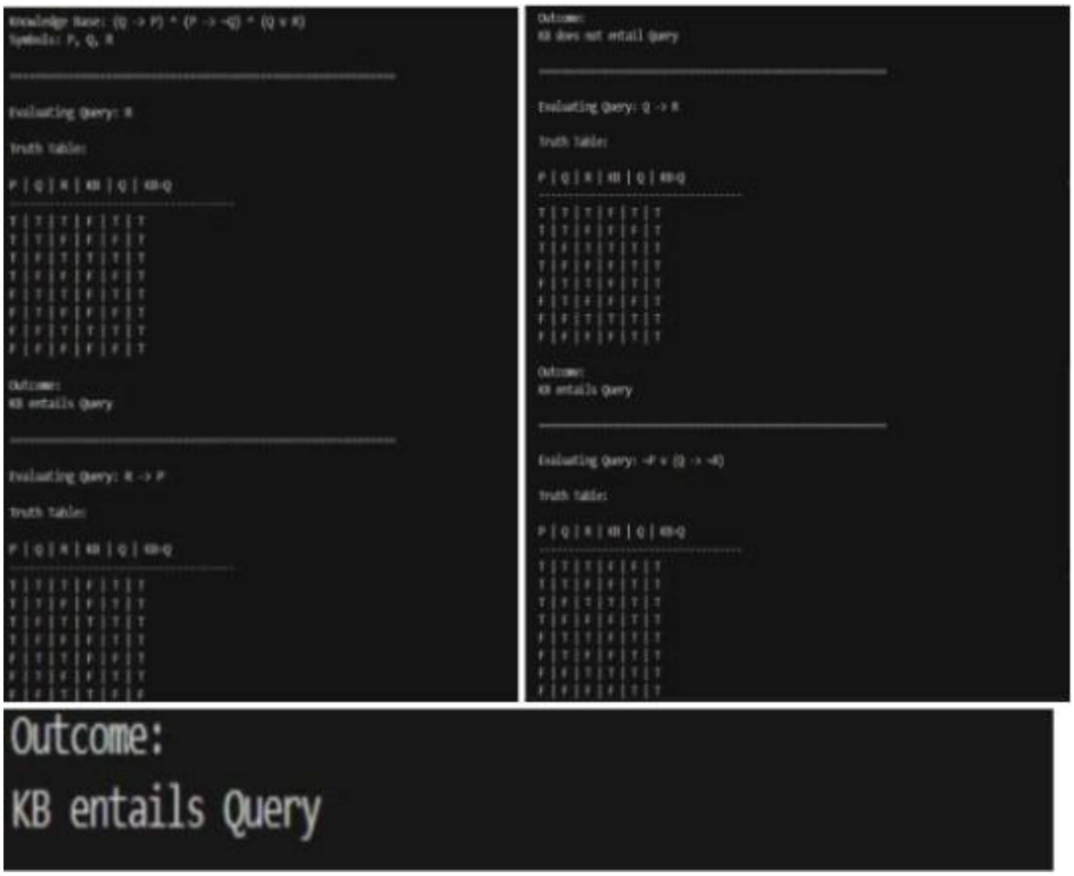
KB ⇒ α : True

CODE:

```
import itertools
import re

def interpret(expr, model):
    expr = expr.replace("<->", " == ")
    expr = expr.replace("->", " <= ")
    expr = re.sub(r'~(\w+)', r'(not \1)', expr)
    expr = re.sub(r'~\(([^)]+)\)', r'(not (\1))', expr)
    expr = expr.replace("^", " and ")
    expr = expr.replace("v", " or ")
    for s, v in model.items():
        expr = re.sub(r'\b' + re.escape(s) + r'\b', str(v), expr)
    return eval(expr)

def truth_table(kb, query, symbols):
    all_models = list(itertools.product([True, False], repeat=len(symbols)))
    ent = True

    print("Truth Table:\n")
    head = " | ".join(symbols) + " | KB | Q | KB⇒Q"
    print(head)
    print("-" * (len(head) + 10))

    for vals in all_models:
        model = dict(zip(symbols, vals))
        kb_v = interpret(kb, model)
        q_v = interpret(query, model)
        impl = (not kb_v) or q_v
```

```
        if kb_v and not q_v:
            ent = False

        row = " | ".join(["T" if v else "F" for v in vals])
        row += f" | {'T' if kb_v else 'F'} | {'T' if q_v else 'F'} | {'T' if impl else 'F'}"
        print(row)

    print("\nOutcome:")
    if ent:
        print("KB entails Query")
    else:
        print("KB does not entail Query")

def run_evaluation(kb, queries, symbols):
    print("\nKnowledge Base:", kb)
    print("Symbols:", ", ".join(symbols))
    print("\n" + "="*60 + "\n")
    for q in queries:
        print("Evaluating Query:", q)
        print()
        truth_table(kb, q, symbols)
        print("\n" + "="*60 + "\n")

kb = "(Q -> P) ^ (P -> ~Q) ^ (Q v R)"
symbols = ["P", "Q", "R"]
queries = ["R", "R -> P", "Q -> R", "~P v (Q -> ~R)"]

run_evaluation(kb, queries, symbols)
```

Output:



```
Knowledge Base: (Q -> P) ^ (P -> ¬Q) ^ (Q v R)
Symbols: P, Q, R

------------------------------------------------------------

Evaluating Query: R

Truth Table:

P | Q | R | KB | Q | KB⇒Q
-----------------------------------
T | T | T | F | T | T
T | T | F | F | F | T
T | F | T | T | T | T
T | F | F | F | F | T
F | T | T | F | T | T
F | T | F | F | F | T
F | F | T | T | T | T
F | F | F | F | F | T

Outcome:
KB entails Query

------------------------------------------------------------

Evaluating Query: R -> P

Truth Table:

P | Q | R | KB | Q | KB⇒Q
-----------------------------------
T | T | T | F | T | T
T | T | F | F | T | T
T | F | T | T | T | T
T | F | F | F | T | T
F | T | T | F | F | T
F | T | F | F | T | T
F | F | T | T | F | F
```

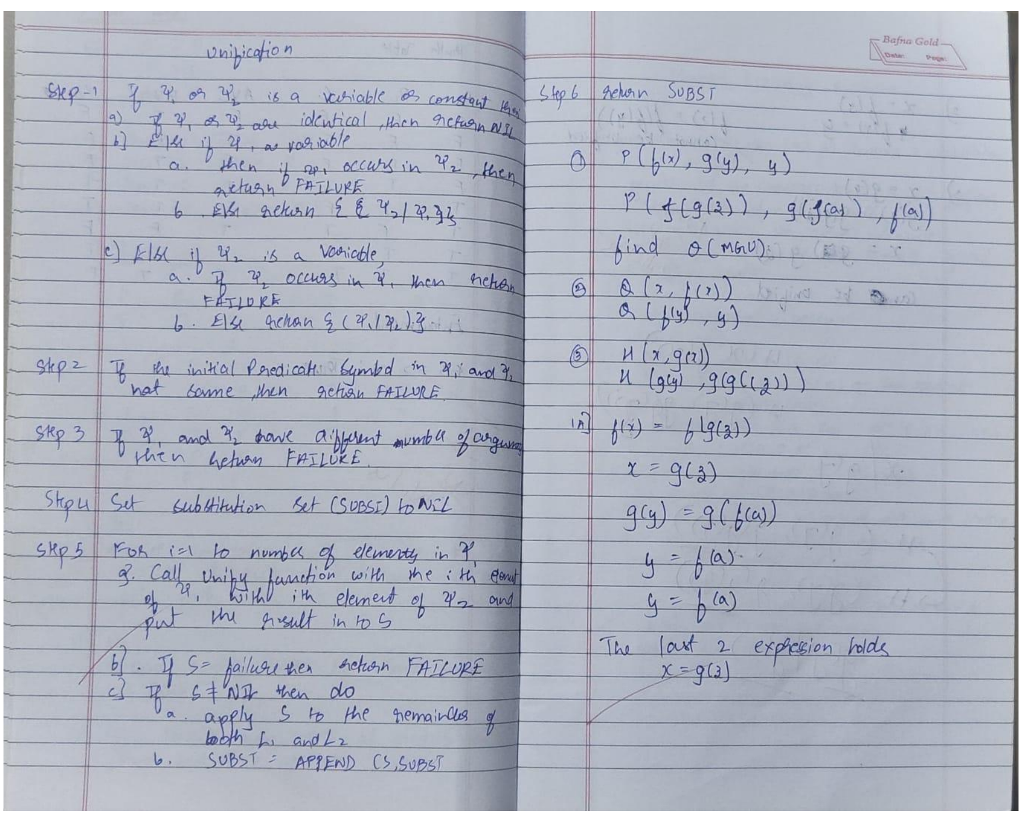
```
Outcome:
KB does not entail Query

------------------------------------------------------------

Evaluating Query: Q -> R

Truth Table:

P | Q | R | KB | Q | KB⇒Q
-----------------------------------
T | T | T | F | T | T
T | T | F | F | F | T
T | F | T | T | T | T
T | F | F | F | T | T
F | T | T | F | T | T
F | T | F | F | F | T
F | F | T | T | T | T
F | F | F | F | T | T

Outcome:
KB entails Query

------------------------------------------------------------

Evaluating Query: ¬P v (Q -> ¬R)

Truth Table:

P | Q | R | KB | Q | KB⇒Q
-----------------------------------
T | T | T | F | F | T
T | T | F | F | T | T
T | F | T | T | T | T
T | F | F | F | T | T
F | T | T | F | T | T
F | T | F | F | T | T
F | F | T | T | T | T
F | F | F | F | T | T
```

```
Outcome:
KB entails Query
```

PROGRAM 7:

Implement unification in first order logic

ALGORITHM:



Unification

Step-1 If Ψ1 or Ψ2 is a variable or constant then
a) If Ψ1 or Ψ2 are identical, then return NIL
b) Else if Ψ1 a variable
a. then if Ψ1 occurs in Ψ2, then return FAILURE
b. Else return {(Ψ2/Ψ1)}

c) Else if Ψ2 is a variable,
a. If Ψ2 occurs in Ψ1 then return FAILURE
b. Else return {(Ψ1/Ψ2)}

Step 2 If the initial Predicate symbol in Ψ1 and Ψ2 not same, then return FAILURE.

Step 3 If Ψ1 and Ψ2 have different number of arguments then return FAILURE.

Step 4 Set substitution set (SUBST) to NIL

Step 5 For i=1 to number of elements in Ψ1
a. Call Unify function with the i th element of Ψ1 with i th element of Ψ2 and put the result in to S
b. If S = failure then return FAILURE
c) If S ≠ NIL then do
a. apply S to the remainder of both L1 and L2
b. SUBST = APPEND (S, SUBST)

Step 6 return SUBST

1) P(f(x), g(y), y)
P(f(g(z)), g(f(a)), f(a))
find θ (MGU)

2) Q(x, f(x))
Q(f(y), y)

3) H(x, g(x))
H(g(y), g(g(z)))

1A] f(x) = f(g(z))
x = g(z)
g(y) = g(f(a))
y = f(a)
y = f(a)
The last 2 expression holds
x = g(z)

CODE:

```
import time
class Term:
    def __init__(self, value):
        self.value = value
    def __eq__(self, other):
        return isinstance(other, Term) and self.value == other.value
    def __hash__(self):
        return hash(self.value)
    def __repr__(self):
        return str(self.value)

class Constant(Term):
    pass

class Variable(Term):
    pass

class Function(Term):
    def __init__(self, symbol, args):
        self.value = symbol
        self.arguments = list(args)
    def __eq__(self, other):
        return isinstance(other, Function) and self.value == other.value and self.arguments ==
other.arguments
    def __hash__(self):
        return hash((self.value, tuple(self.arguments)))
    def __repr__(self):
        return f"{self.value}({', '.join(map(str, self.arguments))})"

FAILURE = "FAILURE"

def occurs_in(v, t):
    if v == t:
        return True
    if isinstance(t, Function):
        return any(occurs_in(v, a) for a in t.arguments)
    return False

def substitute(subst, t):
    if not subst:
        return t
    if isinstance(t, Variable):
        return subst.get(t, t)
    if isinstance(t, Function):
        return Function(t.value, [substitute(subst, a) for a in t.arguments])
    return t

def compose(s1, s2):
    if not s1:
        return s2 or {}
    if not s2:
        return dict(s1)
    r = {v: substitute(s1, t) for v, t in s2.items()}
    for v, t in s1.items():
```

```
        if v not in r:
            r[v] = t
    return r

def unify_core(a, b):
    if a == b:
        return {}
    if isinstance(a, Variable):
        if occurs_in(a, b):
            return FAILURE
        return {a: b}
    if isinstance(b, Variable):
        if occurs_in(b, a):
            return FAILURE
        return {b: a}
    if isinstance(a, Function) and isinstance(b, Function) and a.value == b.value and len(a.arguments)
== len(b.arguments):
        s = {}
        for x, y in zip(a.arguments, b.arguments):
            x2 = substitute(s, x)
            y2 = substitute(s, y)
            r = unify_core(x2, y2)
            if r == FAILURE:
                return FAILURE
            if r:
                s = compose(r, s)
        return s
    return FAILURE

def unify(a, b):
    return unify_core(a, b)

def pretty(s):
    if not s:
        return "{}"
    return "{" + ", ".join([f"{k}->{v}" for k, v in s.items()]) + "}"
def trace_unify(a, b):
    print("Unify:")
    print("A =", a)
    print("B =", b)
    print()
    steps = [(a,
    b)] subst = {}
    step = 1
    while steps:
        x, y = steps.pop(0)
        x = substitute(subst, x)
        y = substitute(subst, y)
        print("Step", step)
        print("Compare:", x, "and", y)
        if x == y:
            print("Equal\n")
            step += 1
            time.sleep(0.1)
```

```
            continue
        if isinstance(x, Variable):
            if occurs_in(x, y):
                print("Failure\n")
                return FAILURE
            s = {x: y}
            subst = compose(s, subst)
            print("Bind:", pretty(s))
            print("Now:", pretty(subst), "\n")
            step += 1
            time.sleep(0.1)
            continue
        if isinstance(y, Variable):
            if occurs_in(y, x):
                print("Failure\n")
                return FAILURE
            s = {y: x}
            subst = compose(s, subst)
            print("Bind:", pretty(s))
            print("Now:", pretty(subst), "\n")
            step += 1
            time.sleep(0.1)
            continue
        if isinstance(x, Function) and isinstance(y, Function) and x.value == y.value:
            for a1, a2 in zip(x.arguments, y.arguments):
                steps.insert(0, (a1, a2))
            print("Expand arguments\n")
            step += 1
            time.sleep(0.1)
            continue
        print("Failure\n")
        return FAILURE
    print("Final substitution:", pretty(subst), "\n")
    return subst

X = Variable("X")
Y = Variable("Y")
A = Constant("a")
B = Constant("b")

t1 = Function("P", [X, A])
t2 = Function("P", [B, Y])

t3 = Function("R", [X, Function("f", [X])])
t4 = Function("R", [A, Function("f", [B])])

print("\n=== Example 1 ===\n")
trace_unify(t1, t2)

print("\n=== Example 2 ===\n")
trace_unify(t3, t4)
```

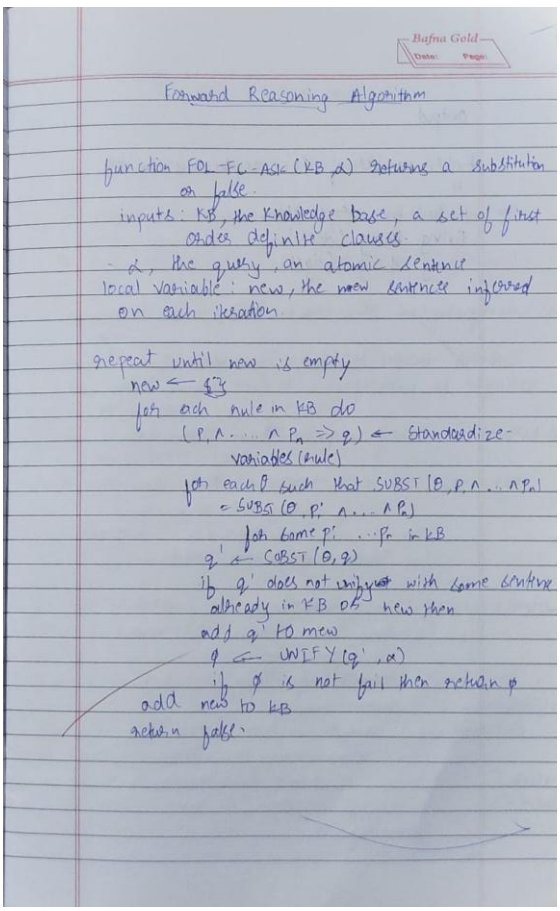
OutPut:

```
=== Unification Algorithm ===
Enter first expression: H(x,g(x))
Enter second expression: H(g(y),g(g(z)))

✅ Unification Successful!
Substitutions:
 x / g(y)
 y / z
```

PROGRAM 8:

Create a knowledge base consisting of first order logic statements and prove the given query using forward reasoning.

ALOGRITHM


Bafna Gold
Date: Page:

Forward Reasoning Algorithm

function FOL-FC-ASK (KB, α) returns a substitution or false.
inputs: KB, the knowledge base, a set of first order definite clauses.
α, the query, an atomic sentence
local variable: new, the new sentence inferred on each iteration.

repeat until new is empty
new ← {}
for each rule in KB do
($P_1 \wedge \ldots \wedge P_n \Rightarrow q$) ← Standardize-variables (rule)
for each θ such that SUBST (θ, $P_1 \wedge \ldots \wedge P_n$) = SUBST (θ, $P_1' \wedge \ldots \wedge P_n'$)
for some $p_1' \ldots p_n'$ in KB
q' ← SUBST (θ, q)
if q' does not unify with some sentence already in KB or new then
add q' to new
φ ← UNIFY (q', α)
if φ is not fail then return φ
add new to KB
return false.

:

CODE:

```
class Predicate:
    def __init__(self, name, args):
        self.name = name
        self.args = args

    def __repr__(self):
        return f"{self.name}({', '.join(map(str, self.args))})"

    def __eq__(self, other):
        return isinstance(other, Predicate) and self.name == other.name and self.args == other.args

    def __hash__(self):
        return hash((self.name, tuple(self.args)))

class Var:
    def __init__(self, name):
        self.name = name

    def __repr__(self): return
        self.name

    def __eq__(self, other):
        return isinstance(other, Var) and self.name == other.name

    def __hash__(self): return
        hash(self.name)

class Const:
    def __init__(self, name):
        self.name = name

    def __repr__(self): return
        self.name

    def __eq__(self, other):
        return isinstance(other, Const) and self.name == other.name

    def __hash__(self): return
        hash(self.name)

def is_variable(x):
    return isinstance(x, Var)

def unify(x, y, subst={}):
    if subst is None:
        return None
    elif x == y:
        return subst
    elif is_variable(x):
        return unify_var(x, y, subst)
    elif is_variable(y):
        return unify_var(y, x, subst)
    elif isinstance(x, Predicate) and isinstance(y, Predicate):
```

```
        if x.name != y.name or len(x.args) != len(y.args):
            return None
        for a, b in zip(x.args, y.args):
            subst = unify(a, b, subst)
            if subst is None:
                return None
        return subst
    else:
        return None

def unify_var(var, x, subst):
    if var in subst:
        return unify(subst[var], x, subst)
    elif is_variable(x) and x in subst:
        return unify(var, subst[x], subst)
    elif occurs_check(var, x, subst):
        return None
    else:
        subst_copy = subst.copy()
        subst_copy[var] = x
        return subst_copy

def occurs_check(var, x, subst):
    if var == x:
        return True
    elif is_variable(x) and x in subst:
        return occurs_check(var, subst[x], subst)
    elif isinstance(x, Predicate):
        return any(occurs_check(var, arg, subst) for arg in x.args)
    else:
        return False

def substitute(predicate, subst):
    new_args = []
    for arg in predicate.args:
        val = arg
        while is_variable(val) and val in subst:
            val = subst[val]
        new_args.append(val)
    return Predicate(predicate.name, new_args)

class Rule:
    def __init__(self, premises, conclusion):
        self.premises = premises
        self.conclusion = conclusion

    def __repr__(self):
        return f"{' ∧ '.join(map(str, self.premises))} => {self.conclusion}"

def forward_chain(kb_facts, kb_rules, query):
    inferred = set(kb_facts)
    print("Initial Facts:")
    for f in inferred:
```

```python
        print(f" {f}")
    print("\nStarting inference steps:\n")

    new_inferred = True

    while new_inferred:
        new_inferred = False
        for rule in kb_rules:
            possible_substs = [{}]

            for premise in rule.premises:
                temp_substs = []
                for fact in inferred:
                    for subst in possible_substs:
                        subst_try = unify(premise, fact, subst)
                        if subst_try is not None:
                            temp_substs.append(subst_try)
                possible_substs = temp_substs

            for subst in possible_substs:
                concluded_fact = substitute(rule.conclusion, subst)
                if concluded_fact not in inferred:
                    print(f"Inferred: {concluded_fact} from rule {rule} using substitution {subst}")
                    inferred.add(concluded_fact)
                    new_inferred = True
                    if unify(concluded_fact, query) is not None:
                        print(f"\nQuery {query} proved!")
                        return True

    print(f"\nQuery {query} not proved.")
    return False

if __name__ == "__main__":
    a = Const('a')
    b = Const('b')
    c = Const('c')
    x = Var('x')
    y = Var('y')
    z = Var('z')

    kb_facts = {
        Predicate('Parent', [a, b]),
        Predicate('Parent', [b, c]),
    }

    kb_rules = [
        Rule([Predicate('Parent', [x, y])], Predicate('Ancestor', [x, y])),
        Rule([Predicate('Parent', [x, y]), Predicate('Ancestor', [y, z])], Predicate('Ancestor', [x, z])),
    ]

    query = Predicate('Ancestor', [a, c])

    print("Running forward chaining...\n")
    forward_chain(kb_facts, kb_rules, query)
```

Output:

```
Running forward chaining...

Initial Facts:
  Parent(a, b)
  Parent(b, c)

Starting inference steps:

Inferred: Ancestor(a, b) from rule Parent(x, y) -> Ancestor(x, y) using substitution {x: a,
 y: b}
Inferred: Ancestor(b, c) from rule Parent(x, y) -> Ancestor(x, y) using substitution {x: b,
 y: c}
Inferred: Ancestor(a, c) from rule Parent(x, y) ∧ Ancestor(y, z) -> Ancestor(x, z) using su
bstitution {x: a, y: b, z: c}

Query Ancestor(a, c) proved!
```

PROGRAM 9:

Create a knowledge base consisting of first order logic statements and prove the given query using Resolution

ALGORITHM:

Bafna Gold
Date: Page:

Resolution in First order Logic

1. Convert all sentences to CNF
2. Negate conclusion S & convert result to CNF
3. Add negated conclusion S to premise clauses
4. Repeat until contradiction or no progress is made:
   a. Select 2 clauses (call them parent clauses)
   b. Resolve them together, performing all required unification.
   c. If resolvent is the empty clause, a contradiction has been found
   d. If not, add resolvent to the premises

Output
John like peanuts

¬ likes (John, Peanuts) ¬ food (x) ∨ likes(John

¬ food(Peanuts) ¬ eats (y, z) ∨ killed(y) ∨ food (z)

killed (Anil) ¬ alive(x) ¬ alive(x) ∨ ¬ killed

{} Hence Proved

CODE:

```
def is_variable(x):
return x[0].islower() and len(x) == 1

def unify(x, y, subs=None):
    if subs is None:
        subs = {}
    if x == y:
        return subs
    if is_variable(x):
        return unify_var(x, y, subs)
    if is_variable(y):
        return unify_var(y, x, subs)
    if isinstance(x, list) and isinstance(y, list) and len(x) == len(y):
        for a, b in zip(x, y):
            subs = unify(a, b, subs)
            if subs is None:
                return None
        return subs
    return None

def unify_var(var, x, subs):
    if var in subs:
        return unify(subs[var], x, subs)
    if x in subs:
        return unify(var, subs[x], subs)
    if occurs_check(var, x, subs):
        return None
    subs[var] = x
    return subs

def occurs_check(var, x, subs):
    if var == x:
        return True
    if isinstance(x, list):
        return any(occurs_check(var, xi, subs) for xi in x)
    if x in subs:
        return occurs_check(var, subs[x], subs)
    return False

def negate(literal):
    return literal[1:] if literal.startswith("¬") else "¬" + literal
```

```
def parse_predicate(pred):
    name, args = pred.split("(")
    args = args[:-1].split(",")
    args = [a.strip() for a in args]
    return name.strip(), args

def substitute(literal, subs):
    name, args = parse_predicate(literal.replace("¬", ""))
    new_args = [subs.get(a, a) for a in args]
    new_lit = name + "(" + ", ".join(new_args) + ")" return
    "¬" + new_lit if literal.startswith("¬") else new_lit

def unify_predicates(p1, p2):
    p1_clean = p1.replace("¬", "")
    p2_clean = p2.replace("¬", "")
    n1, a1 = parse_predicate(p1_clean)
    n2, a2 = parse_predicate(p2_clean)
    if n1 != n2 or len(a1) != len(a2):
        return         None
    return unify(a1, a2)

def resolve(ci, cj):
    for li in ci:
        for lj in cj:
            if lj == negate(li):
                subs = unify_predicates(li, lj)
                if subs is not None:
                    new_clause = set()
                    for l in ci.union(cj):
                        if l != li and l != lj:
                            new_clause.add(substitute(l, subs))
                    return new_clause, (li, lj), subs
    return None, None, None
```

```
KB = [
    {"¬food(x)", "likes(John,x)"},
    {"food(Apple)"},
    {"food(vegetables)"},
    {"¬eats(y,z)", "killed(y)", "food(z)"},
    {"eats(Anil,Peanuts)"},
    {"alive(Anil)"},
    {"¬eats(Anil,w)", "eats(Harry,w)"},
    {"killed(g)", "alive(g)"},
    {"¬alive(k)", "¬killed(k)"},
    {"likes(John,Peanuts)"}
]

def resolution_tree(KB, query):

    print(" PROOF BY RESOLUTION (FOL) ")
    print(f"Goal: Prove that {query}\n")
    print("Converting to CNF and negating the query...\n")

    clauses = [c.copy() for c in KB]
    negated_query = negate(query)
    clauses.append({negated_query})
    print(f"Negated query added to KB: {negated_query}\n")
    new = []
    step = 1
    proof_steps = []
```

```
    tree_nodes = []

    while True:
        pairs = [(clauses[i], clauses[j]) for i in range(len(clauses))
                 for j in range(i + 1, len(clauses))]

        for (ci, cj) in pairs:
            resolvent, used, subs = resolve(ci, cj)
            if resolvent is not None:
                proof_steps.append({
                    "step": step,
                    "parents": (ci, cj),
                    "used": used,
                    "subs": subs,
                    "resolvent": resolvent
                })

                print(f"Step {step}: Resolving {ci} and {cj}")
                print(f" Used literals: {used}")
                if subs:
                    print(f" Substitution: {subs}")
                print(f" ⇒ New Clause:
                 {resolvent}\n")
                tree_nodes.append((ci, cj, resolvent))


                if not resolvent:


                    print(" Empty clause derived — Query is PROVED!\n")
                    print("\n============ PROOF TREE ============\n")
                    print_tree(tree_nodes)
                    return True

                if resolvent not in clauses:
                    new.append(resolvent)
                step += 1
        if not new:
            print(" No new clauses can be derived — Query cannot be proven.\n")
            print("\n============ PROOF TREE (Partial) ============\n")
            print_tree(tree_nodes)

            return False

        clauses.extend(new)
        new = []

def print_tree(nodes):
    def print_branch(node, level=0):
        indent = " " * level
        c1, c2, result = node
        print(f"{indent} ├─ Derived {result} from:")
        print(f"{indent} │  {c1}")
        print(f"{indent} │  {c2}")
```

```
    for n in nodes:
        print_branch(n)
        print("")

if name == " main ": query
    = "likes(John,Peanuts)"
    result = resolution_tree(KB, query)
    print("Result:", " PROVED" if result else " NOT PROVED")
```

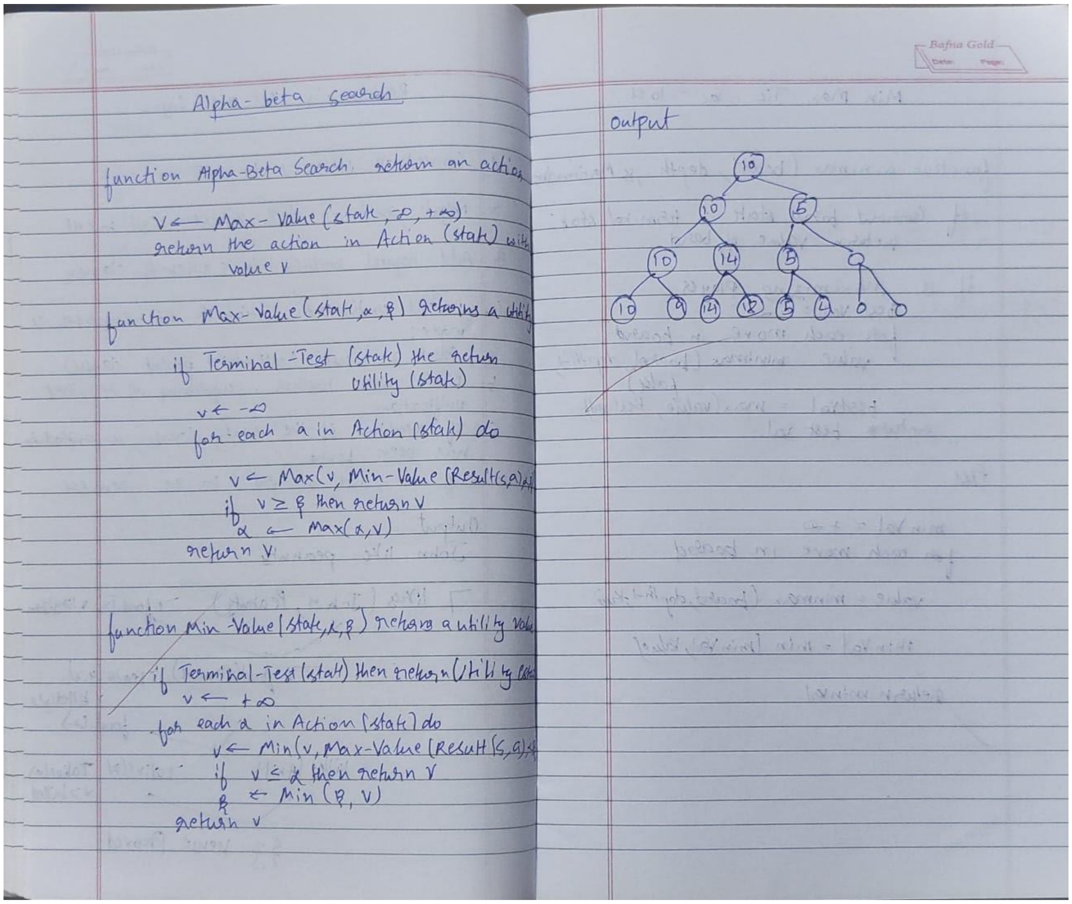
OutPut:

```
        PROOF BY RESOLUTION (FOL)
Goal: Prove that likes(John,Peanuts)

Converting to CNF and negating the query...

Negated query added to KB: ¬likes(John,Peanuts)

Step 1: Resolving {'likes(John,Peanuts)'} and {'¬likes(John,Peanuts)'}
       Used literals: ('likes(John,Peanuts)', '¬likes(John,Peanuts)')
       ⇒ New Clause: set()

 Empty clause derived — Query is PROVED!


============ PROOF TREE ============

├─ Derived set() from:
│    {'likes(John,Peanuts)'}
│    {'¬likes(John,Peanuts)'}

Result:  PROVED
```

PROGRAM 10:

Implement Alpha-Beta Pruning.

ALGORITHM:



Alpha- beta search

function Alpha-Beta Search return an action

$v \leftarrow$ Max-Value (state, $-\infty$, $+\infty$)
return the action in Action (state) with value v

function Max-Value (state, α, β) returns a utility

if Terminal-Test (state) the return utility (state)
$v \leftarrow -\infty$
for each a in Action (state) do
$v \leftarrow$ Max(v, Min-Value (Result(s,a),α,β
if $v \geq \beta$ then return v
$\alpha \leftarrow$ Max(α, v)
return v

function Min-Value (state, α, β) returns a utility value

if Terminal-Test (state) then return Utility (state
$v \leftarrow +\infty$
for each a in Action (state) do
$v \leftarrow$ Min(v, Max-Value (Result (s, a),α,β
if $v \leq \alpha$ then return v
$\beta \leftarrow$ Min (β, v)
return v

Output

CODE:

```
import math


tree = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': ['H', 'I'],
    'E': ['J', 'K'],
    'F': ['L', 'M'],
    'G': ['N', 'O'],
    'H': [], 'I': [], 'J': [], 'K': [],
    'L': [], 'M': [], 'N': [], 'O': []
}

values = {
    'H': 3, 'I':
    5,
    'J': 6, 'K': 9,
    'L': 1, 'M': 2,
    'N': 0, 'O': -1
}

def get_children(node):
```

```
    return tree.get(node, [])

def evaluate(node):
    return values.get(node, 0)

def alphabeta(node, depth, alpha, beta, maximizingPlayer):
    if not get_children(node) or depth == 0:
        return evaluate(node)

    if maximizingPlayer:
        value = -math.inf
        print(f"MAX Node {node}: α={alpha}, β={beta}")
        for child in get_children(node):
            value = max(value, alphabeta(child, depth - 1, alpha, beta, False))
            alpha = max(alpha, value)
            print(f" MAX updating α={alpha} after visiting {child}")
            if alpha >= beta:
                print(f" Pruned remaining children of {node} (α={alpha}, β={beta})")
                break
        return value
    else:
        value = math.inf
        print(f"MIN Node {node}: α={alpha}, β={beta}")
        for child in get_children(node):
            value = min(value, alphabeta(child, depth - 1, alpha, beta, True))
            beta = min(beta, value)
            print(f" MIN updating β={beta} after visiting {child}")
            if beta <= alpha:
                print(f" Pruned remaining children of {node} (α={alpha}, β={beta})")
                break
        return value


print("\n--- Running Alpha–Beta Pruning on Minimax Tree ---\n")
optimal_value = alphabeta('A', depth=4, alpha=-math.inf, beta=math.inf, maximizingPlayer=True)
print("\nOptimal value at the root (A):", optimal_value)
```

OutPut:

```
--- Running Alpha-Beta Pruning on Minimax Tree ---

MAX Node A: α=-inf, β=inf
MIN Node B: α=-inf, β=inf
MAX Node D: α=-inf, β=inf
  MAX updating α=3 after visiting H
  MAX updating α=5 after visiting I
  MIN updating β=5 after visiting D
MAX Node E: α=-inf, β=5
  MAX updating α=6 after visiting J
   Pruned remaining children of E (α=6, β=5)
  MIN updating β=5 after visiting E
  MAX updating α=5 after visiting B
MIN Node C: α=5, β=inf
MAX Node F: α=5, β=inf
  MAX updating α=5 after visiting L
  MAX updating α=5 after visiting M
  MIN updating β=2 after visiting F
   Pruned remaining children of C (α=5, β=2)
  MAX updating α=5 after visiting C

Optimal value at the root (A): 5
```